WORKSHEET 6.13

Unit 6 Synthesis: The Accumulation Free-Response Question

Unit 6: Integration and Accumulation of Change

Key ideas

- Amount at time t = starting amount + $\displaystyle\int_0^t \text{rate}\,ds$.
- The derivative of an accumulation function is its integrand, so its extrema are where the integrand changes sign.
- On a closed interval, compare the values at critical points **and** endpoints.
- Write the integral before evaluating it: the set-up earns a point of its own.

Worked examples

Example 1 · An amount from a rate

A tank holds 20 litres at $t = 0$, and water flows in at the net rate $6 - 2t$ litres per minute. Find the amount at $t = 4$ and the greatest amount for $0 \le t \le 4$.

Solution. $A(t) = 20 + \displaystyle\int_0^t (6 - 2s)\,ds = 20 + 6t - t^2$. $A'(t) = 6 - 2t = 0$ at $t = 3$.

$A(4) = 28$**; the greatest amount is** $A(3) = 29$ **litres.**

Example 2 · Differentiating an accumulation

Find $\dfrac{d}{dx}\displaystyle\int_1^x te^t\,dt$.

Solution. Fundamental Theorem, Part 1.

xe^x**.**

Example 3 · A definite substitution

Evaluate $\displaystyle\int_0^1 6x\left(x^2+1\right)^2\,dx$.

Solution. $u = x^2 + 1$ from 1 to 2: $\displaystyle\int_1^2 3u^2\,du$.

$8 - 1 = 7$**.**

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Practice

CORE

1. Evaluate $\int_0^3 (2x+1)\,dx$.

2. Find $\int \sec^2(2x)\,dx$.

3. Find $\dfrac{d}{dx}\int_0^x \sqrt{t^2+9}\,dt$ and its value at $x=4$.

4. A differentiable function f has $f(1)=3$ and $f(4)=-2$. Find $\int_1^4 f'(x)\,dx$.

EXAM LEVEL

5. Water flows into a pond at the net rate $R(t)=12-3t$ litres per minute for $0\le t\le 6$. Find the net change in the amount of water over the 6 minutes, and the time at which the amount is greatest.

6. Evaluate $\int_0^2 x\sqrt{4-x^2}\,dx$.

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7. Let $F(x) = \int_0^x (t-2)e^t\,dt$. At what value of x does F have a relative minimum? Justify your answer.

8. A particle has velocity $v(t) = t^2 - 4$ for $0 \le t \le 3$. Find the total distance it travels.

Question 9 · Free-response practice

Let $G(x) = \int_0^x \left(3t^2 - 12\right)\,dt$ for $0 \le x \le 4$.

(a) Find $G'(x)$ and $G''(x)$.

(b) Find the x-coordinate of the relative minimum of G, justify it, and find the value of G there.

(c) Find the absolute maximum value of G on $[0, 4]$. Justify your answer.

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Worked solutions

1. $\Big[x^2 + x\Big]_0^3 = 12$.

2. $\dfrac{1}{2}\tan(2x) + C$.

3. $\sqrt{x^2+9}$; at $x = 4$ it is 5.

4. $f(4) - f(1) = -5$.

5. $\displaystyle\int_0^6 (12 - 3t)\,dt = 72 - 54 = 18$ litres. R changes from positive to negative at $t = 4$, so the amount is greatest then.

6. $u = 4 - x^2$ from 4 to 0, $x\,dx = -\dfrac{1}{2}du$: $\dfrac{1}{2}\displaystyle\int_0^4 \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}(8) = \frac{8}{3}$.

7. $F'(x) = (x-2)e^x$ changes from negative to positive at $x = 2$.

8. v changes sign at $t = 2$. $\displaystyle\int_0^2 (t^2-4)\,dt = -\frac{16}{3}$ and $\displaystyle\int_2^3 (t^2-4)\,dt = \frac{7}{3}$. Distance $\dfrac{16}{3} + \dfrac{7}{3} = \dfrac{23}{3}$.

9. (a) $G'(x) = 3x^2 - 12$ and $G''(x) = 6x$.

   (b) $G'(x) = 3(x-2)(x+2)$ changes from negative to positive at $x = 2$. $G(2) = \Big[t^3 - 12t\Big]_0^2 = -16$.

   (c) Candidates: $G(0) = 0$, $G(2) = -16$, $G(4) = 64 - 48 = 16$. The absolute maximum is 16, at $x = 4$.

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